Def. Let G be a Group. Define the upper central series $\{Z_n(G)\}_{n=0}^{\infty}$ as:

$$Z_0(G)=1,$$
$$Z_1(G)=Z(G),$$
$$\vdots$$
$$Z_{i+1}(G) \overset{def}{=} \varphi_i^{-1}[Z(G/Z_i(G))] \quad \text{where } \varphi_i : G \to G/Z_i(G) : x \mapsto xZ_i(G)$$

Note: $1=Z_0(G)\le Z_1(G)\le Z_2(G)\le \cdots$

because $Z_i(G)\le \varphi_i^{-1}[Z(G/Z_i(G))]$ given by

$$x\in Z_i(G) \Rightarrow \varphi_i(x)=xZ_i(G)=1\cdot Z_i(G)\in Z(G/Z_i(G))$$
$$\Rightarrow x\in \varphi_i^{-1}[\varphi_i(x)]\subseteq \varphi_i^{-1}[Z(G/Z_i(G))]$$

and, preimage of Normal is normal, the definition is well-defined.

Clearly, $Z_{i+1}(G)/Z_i(G)=Z(G/Z_i(G))$

$$Z_{i+1}(G)/Z_i(G)=\varphi_i^{-1}[Z(G/Z_i(G))]/Z_i(G) \quad \varphi_i \text{ onto}$$
$$=\varphi_i[\varphi_i^{-1}[Z(G/Z_i(G))]]=Z(G/Z_i(G))$$

Def. A Group G is called Nilpotent if: for some $c\ge 0$, $Z_c(G)=G$.
The smallest such $c\ge 0$ is called the nilpotent class of G.

Prop. Let P be a group of order p^a, where p is prime and $a\ge 1$.
Then P is Nilpotent of nilpotent class at most $a-1$.

Proof. Since p-group has no trivial center, for all $i\ge 0$,

$$|P/Z_i(P)|>1 \Rightarrow Z(P/Z_i(P))\ne 1$$

Therefore, $Z_i(P)\ne P \Rightarrow |P/Z_i(P)|>1 \Rightarrow Z(P/Z_i(P))\ne 1 \Leftrightarrow Z_{i+1}(P)/Z_i(P)>1$

This implies $|Z_{i+1}(P)|\ge p\cdot|Z_i(P)|$, further, $|Z_{i+1}(P)|\ge p^{i+1}$

In particular, $|Z_a(P)|\ge p^a$, at least $i=a$, $Z_a(P)=P$.

If nilpotent class of P is exactly a, then for all $i\ge 0$, $|Z_i(P)|=p^i$.

Then, $P/Z_{a-2}(P)$ has order p^2, so is abelian. Now,

$$Z_{a-1}(P)=\varphi_{a-2}^{-1}[Z(P/Z_{a-2}(P))]=P$$

which is Contradiction.

upper, lower, Central series.